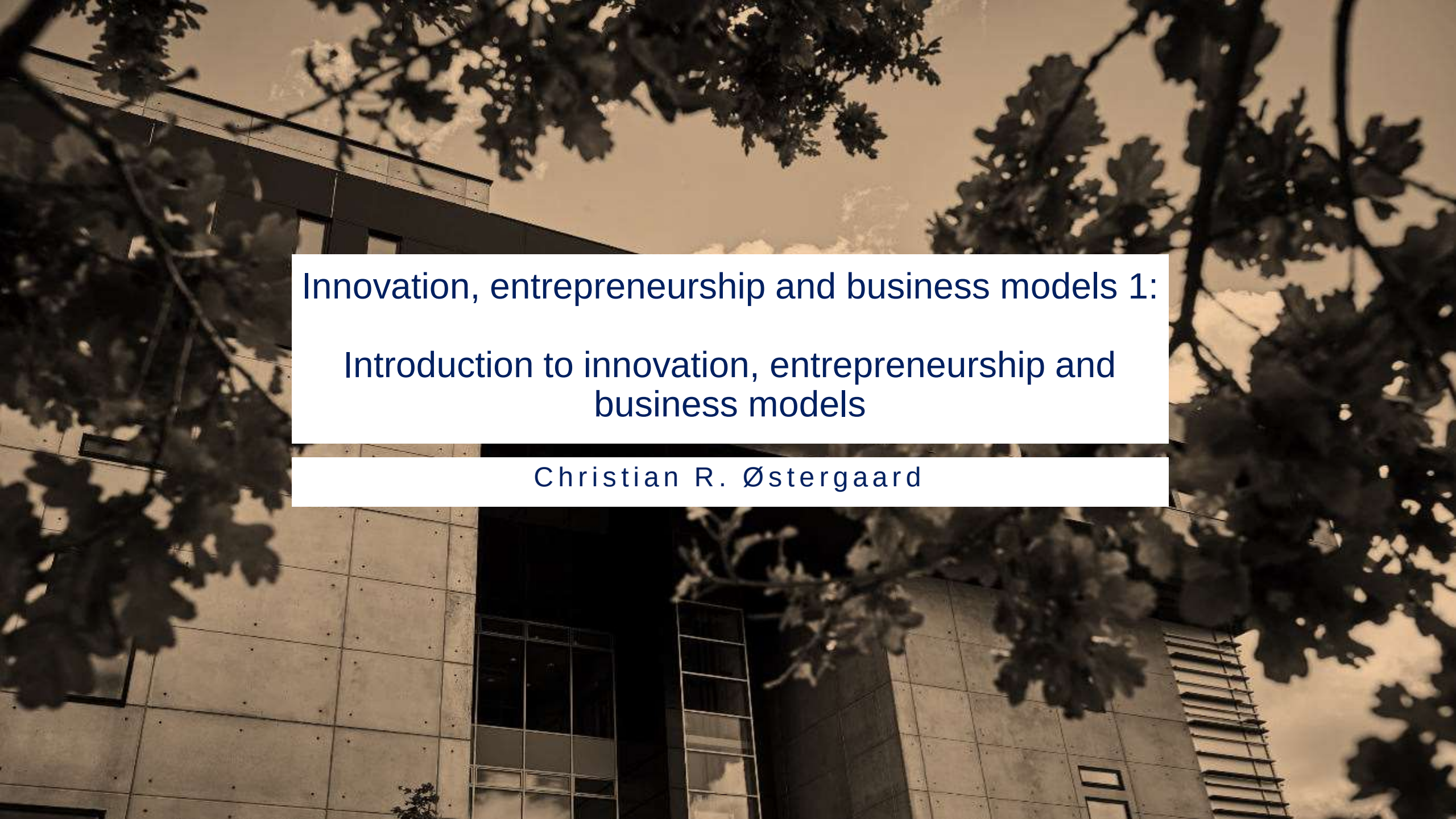


Innovation, entrepreneurship and business models 1:

Introduction to innovation, entrepreneurship and business models

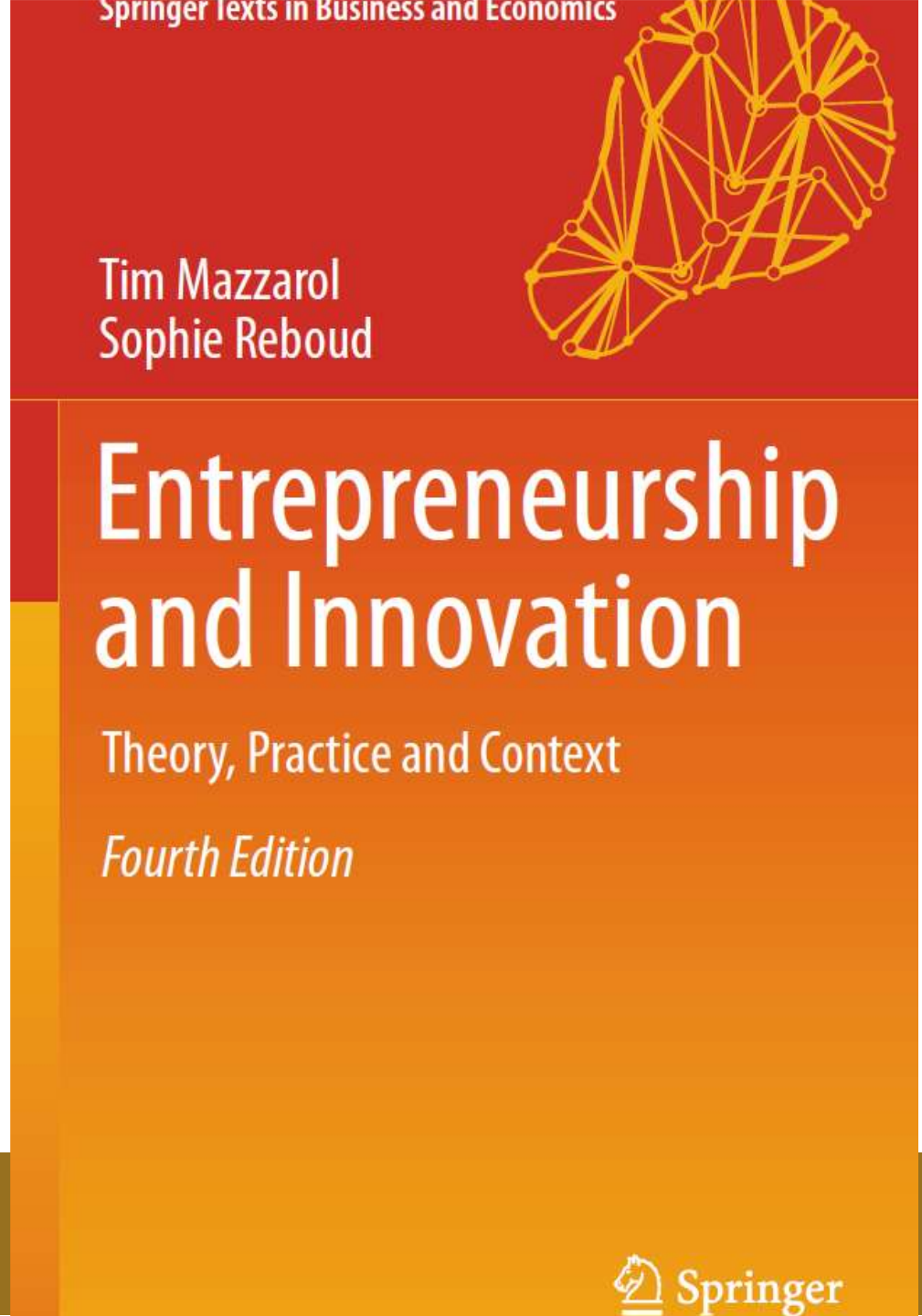
Christian R. Østergaard

Today

- The introductory lecture will address the **structure** of the course, outline the **learning objectives**, introduce the **course lecturers** and the **exam**.
- Then it will discuss the following questions: what is innovation?; what is entrepreneurship?; what are business models?; and why should we care about these terms and their practical application?
- Furthermore, the lecture will focus on who is the innovator, innovation types, effects of innovation and the innovation dilemma.
- The course lecturers

How to approach academic texts for innovation, entrepreneurship and business models

- Why this book?
  - Comprehensive coverage of all relevant topics
  - Online availability for free access
  - Encompasses valuable insights, yet some extraneous information (noise)
  - Note: Authors tend to include numerous models and repetitions
- How to Prepare
  - Focus on reading, not memorization
  - The course is non-linear; avoid expecting a progression from A to B to C
  - Embrace cumulative learning
  - Extract key points and models on entrepreneurship, innovation, and business models



Learning goals

KNOWLEDGE

The objective is that the student after the module possesses the necessary knowledge on:

- key notions, concepts and theories about innovation, entrepreneurship and business models.
- key issues within innovation, entrepreneurship and business models and the ability to identify, define and reflect on these.
- the actions taken by various companies in relation to innovation, entrepreneurship and business models, exemplifying these.

SKILLS

The objective is that the student after the module possesses the necessary skills in:

- identifying theoretical and practical issues within innovation, entrepreneurship and business models.
- applying relevant theories in relation to understanding and analysing a practical issue within innovation, entrepreneurship and business models.
- relating analytically and critically to issues concerning the importance of entrepreneurship and innovation in a dynamic market economy and to businesses in such an economy.
- **COMPETENCES**

The objective is that the student after the module possesses the necessary competences in:

- accepting responsibility for own learning and professional development within innovation, entrepreneurship and business models.
- independently planning and performing analyses of relevant issues within innovation, entrepreneurship and business models.
- combining theory and practice to address issues concerning organisation, management and implementation of innovation and new businesses models.

- Lecture 1: **introduction** to innovation, entrepreneurship and business models (2h lecture),CRO
- Lecture 2: The **entrepreneur** and entrepreneurial processes (2h lecture) JLC
- Lecture 3: **Innovation** in small and large firms (2h lecture) CRO
- Lecture 4: **Entrepreneurship** in practice – the new venture creation (2h lecture) JLC
- Lecture 5: The context for **innovation**, diffusion and adoption (2h lecture +2h exercises) CRO
- Lecture 6: Sourcing in resources for **new venture creation** (2h lecture+ 2h exercises) JLC
- Lecture 7: Disruptive **innovation** (2h lecture + 2h exercises) CRO
- Lecture 8: Diversity in **entrepreneurship** (2h lecture+ 2h exercises) JLC
- Lecture 9: **Business modelling** (2h lecture + 2h exercises) YT
- Lecture 10: Dynamic capabilities and management of risks through **business model** innovation process (2h lecture) YT
- Lecture 11: Recent trends and problems within **entrepreneurship, innovation, business models**. (2h lecture) CRO & YT & JLC

Examination

- The exam is a written individual examination. The exam will be multiple choice based on a set of questions related to issues discussed at the course. The course material is sufficient for answering the questions. The examination will be conducted in Digital Exam and will last for 1.5 hours.

- Date: 31/03-2026
- Place: Skansen
  Lerumbakken 11, 9400 Nørresundby
- Time: 10.00-11.30
- More detailed information will be given later.
- Allowed aids: including textbooks, teaching slides, notes from teaching. It is not allowed to communicate with others or use ChatGPT or other generate AI during the examination.
- Do we need to prepare for the exam?

example

Page 2

According to the definition of innovation by OECD (2018), what are key requirements for an innovation?

Choose one answer

- It must generate significant economic profits.
- It must be novel and unique.
- It must be implemented and put into active use.
- It must be patented by the firm.

← 1 2 3 4 5 6 7 8 9 10 →

0/32

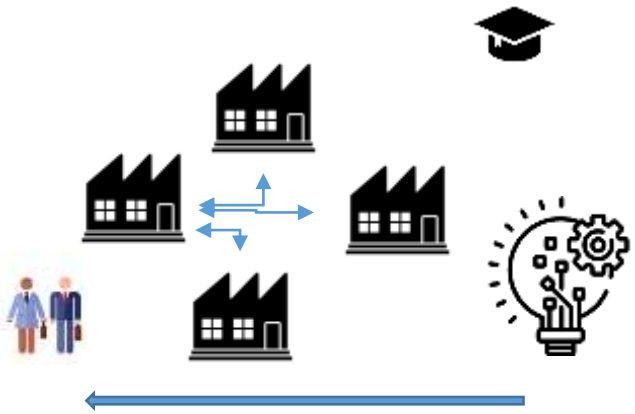
Submit | View results | Remove all answers | Back to the overview

Background

- Innovation, entrepreneurship and business models are key concepts that are necessary to understand changes in and for companies.
- Innovation and entrepreneurship are not only important on a firm-level but are recognized as being among the key elements in the process of economic development and vital to the ability of a nation's economy to maintain ***competitiveness***.
- to maintain their **competitiveness**, companies need to understand these concepts and be able to address the issues and challenges they present.

Innovation and interdependencies

- Innovation and entrepreneurship are both processes and it sometimes requires a broader view (outside –in, inside-out)



- Today we will talk about Chapter 1 and the OECD/Eurostat (2018), Oslo Manual chapter 3

For innovation..

- We care about how innovation influences
  - Growth
  - Competition
  - Job creation - destruction
  - Technological change

- And related to innovation
  - Research and development
  - Patents and IPR protection
  - Human capital /education
  - Knowledge flows
  - Adoption of innovation

Why innovation matte:

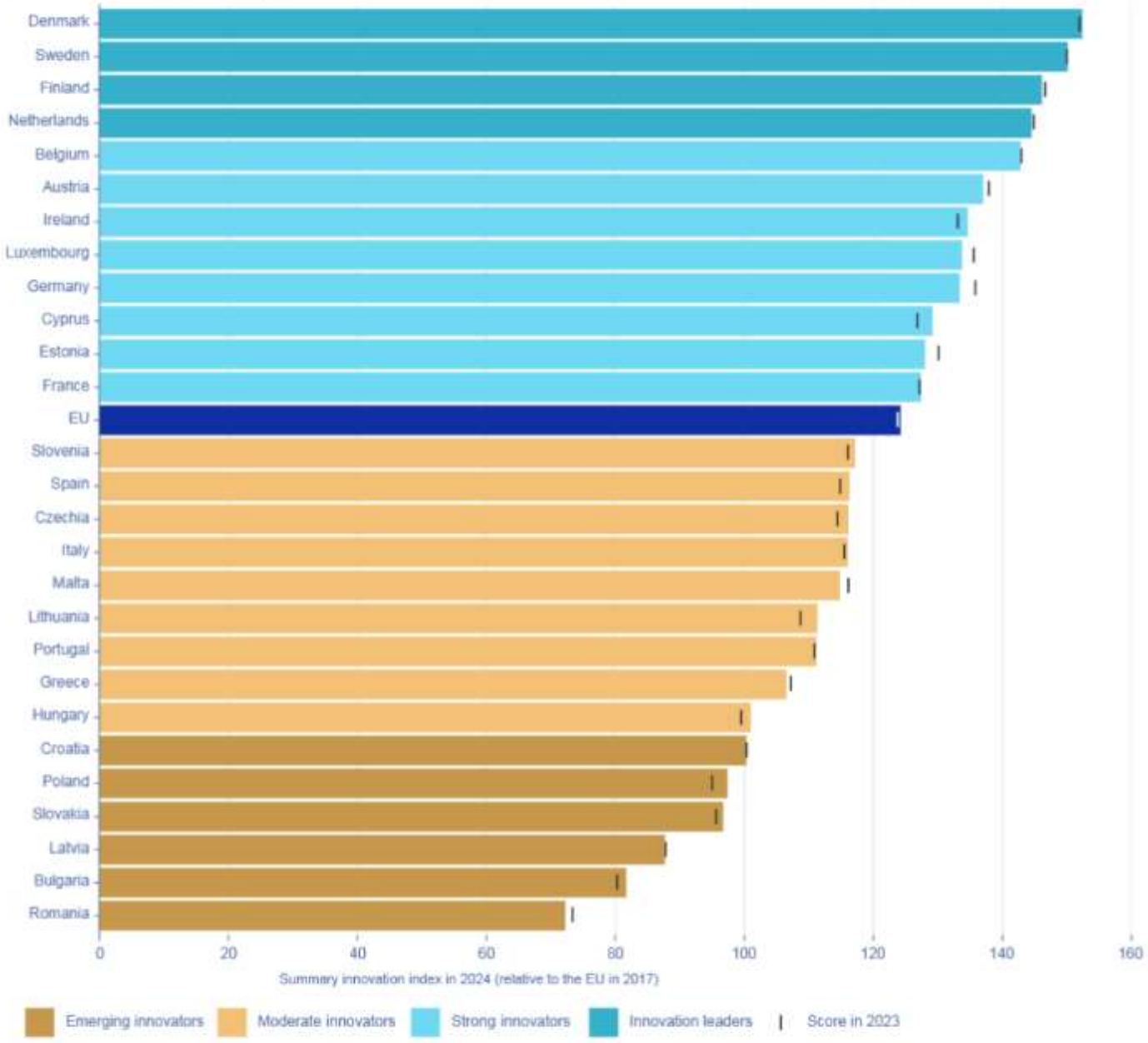
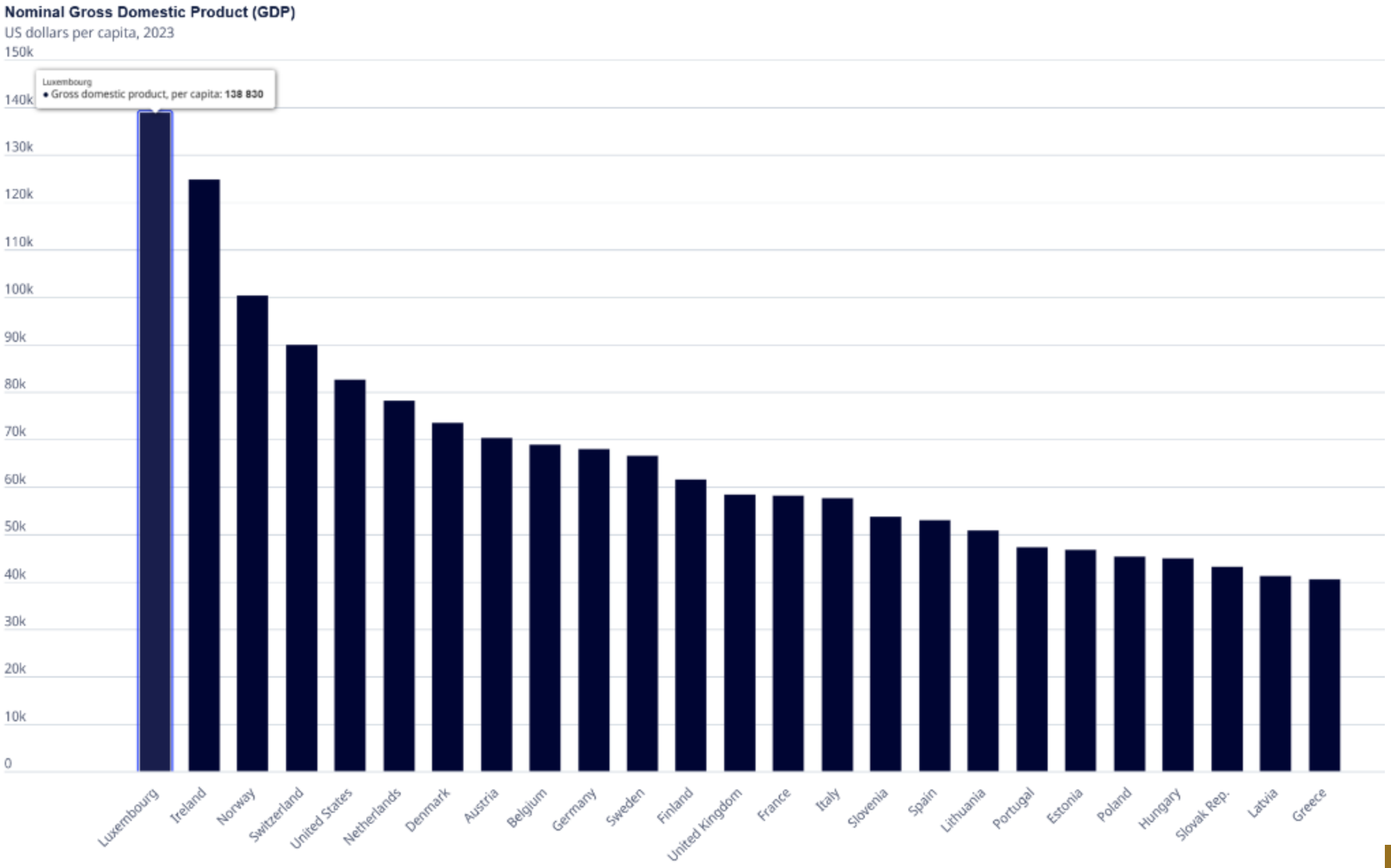


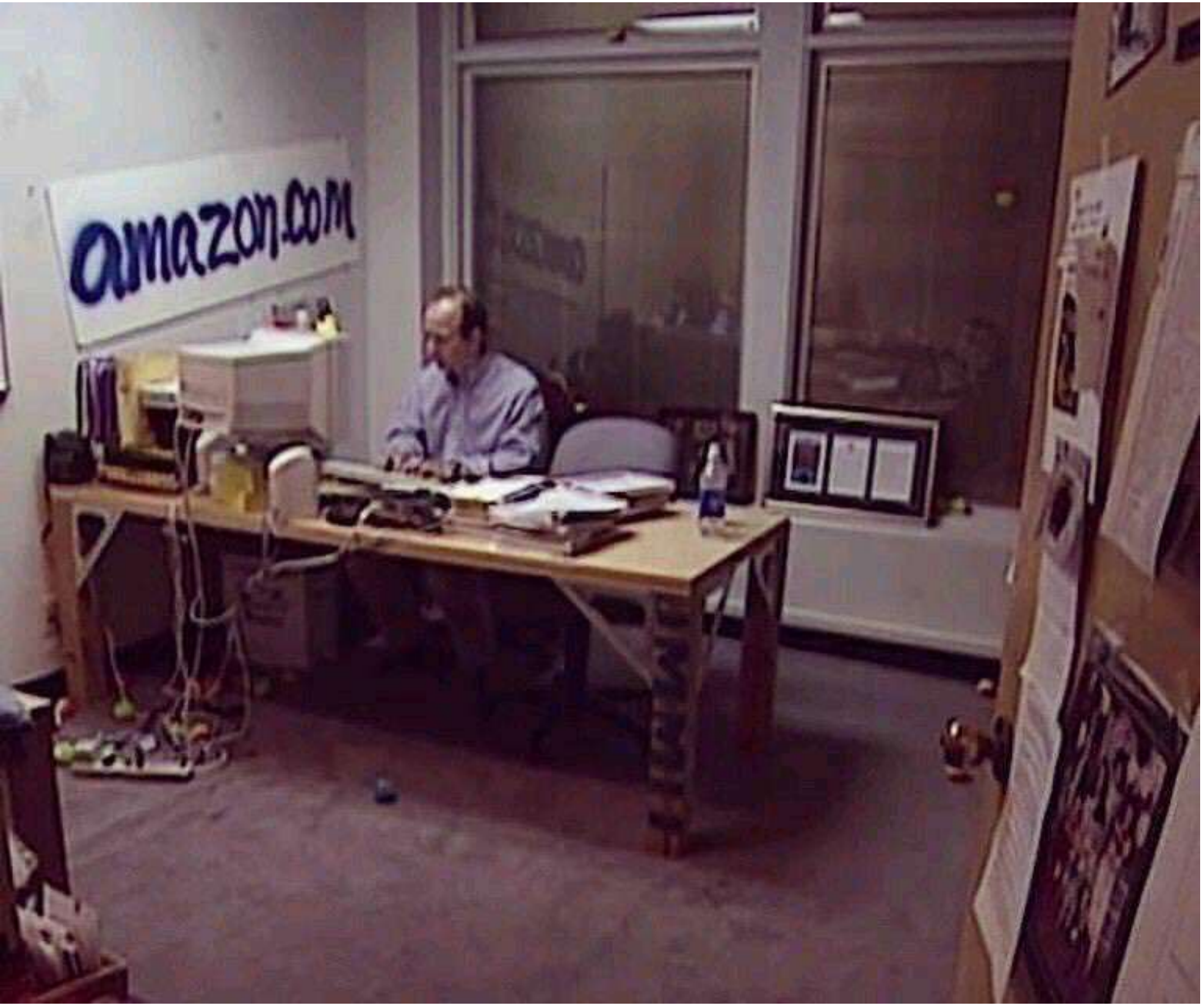
Figure 2: Innovation performance of the EU Member States, relative to the EU in 2017 and compared to national performance in 2023.

Note: All performance scores are relative to that of the EU in 2017. Coloured bars show countries' performance in 2024, using the most recent data for 32 indicators. The vertical bars show performance in 2023, using the next most recent data.


Nominal Gross Domestic Product (GDP)
US dollars per capita, 2023
Luxembourg
Gross domestic product, per capita: 138 830
150k
140k
130k
120k
110k
100k
90k
80k
70k
60k
50k
40k
30k
20k
10k
0
Luxembourg
Ireland
Norway
Switzerland
United States
Netherlands
Denmark
Austria
Belgium
Germany
Sweden
Finland
United Kingdom
France
Italy
Slovenia
Spain
Lithuania
Portugal
Estonia
Poland
Hungary
Slovak Rep.
Latvia
Greece

Problems with entreprenuership and innovation

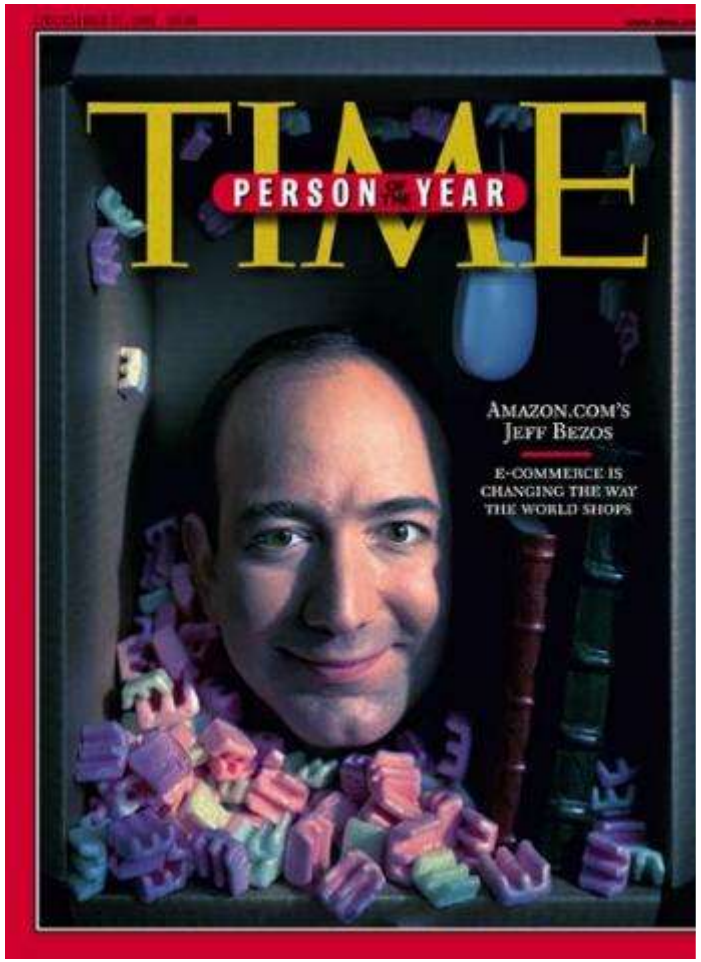
- Conceptual issues
  - Entrepreneurship **create** and **destroy** jobs
  - Innovation **create** and **destroy** jobs
  - Focus on **success -** Google, Apple, Microsoft, Open Ai
- Job creation?
  - SMEs create jobs is technically true, but misleading
    - How many and which jobs
    - Job creation and destuction = churn : entry and exit are correlated
  - Given that **small firms** tend to be **less productive**, have **lower remuneration** (i.e., more volatile), **fewer benefits** (such as pension schemes), **fewer training opportunities**, and more **work-related accidents** (Storey, 1994), increasing SME employment, while large firm employment declines and output remains unchanged, may just be a societal wealth transfer.



Innovation (and entrepreneurship) is also important at the firm level

- Photo of Jeff Bezos in 1999. “Think about this the next time you think you can't do something great”

Same year



Forbes List of World's Richest People, 1999

NEW YORK (AP) -- The ranking of the world's richest people as estimated by *Forbes* magazine, excluding dictators and royalty. Listings include name, home country, wealth in billions of dollars and source of wealth, where available.

1. Gates, William H. III, United States, 90.0, Microsoft
2. Buffett, Warren, United States, 36.0, Berkshire Hathaway
3. Allen, Paul, United States, 30.0, Microsoft
4. Ballmer, Steven, United States, 19.5, Microsoft
5. Oeri, Hoffman and Sacher families, Switzerland, 17.0, Roche pharmaceuticals
6. Anschutz, Philip, United States, 16.5, oil, railroads, telecommunications
7. Dell, Michael, United States, 16.5, Dell Computer
8. Walton, Alice L., United States, 16.0, inheritance (Wal-Mart Stores)
9. Walton, Helen R., United States, 16.0, inheritance (Wal-Mart Stores)
10. Walton, Jim C., United States, 16.0, inheritance (Wal-Mart Stores)
11. Walton, John T., United States, 16.0, inheritance (Wal-Mart Stores)
12. Walton, S. Robson, United States, 15.8, Wal-Mart Stores
13. Alsaud, Prince Alwaleed Bin Talal, Saudi Arabia, 15.0, investments
14. Bettencourt, Liliane, France, 13.9, cosmetics
15. Albrecht, Theo and Karl and family, Germany, 13.6, retailing
16. Li Ka-shing, Hong Kong, 12.7, real estate,
17. Haniel family, Germany, 12.4, pharmaceuticals, diversified
18. Engelhorn, Curt and family, Germany, 12.0, pharmaceuticals
19. Thomson, Kenneth, Canada, 11.9, Thomson Corp
20. Lee Shau Kee, Hong Kong, 11.1, real estate
21. Moore, Gordon Earl, United States, 10.9, Intel
22. Anthony, Barbara Cox, United States, 10.6, inheritance (Cox Enterprises)
23. Chambers, Anne Cox, United States, 10.6, inheritance (Cox Enterprises)
24. Kluge, John, United States, 10.5, Metromedia Co.
25. Walker, Jay, United States, 10.2, Priceline.com
26. Bezos, Jeffrey, United States, 10.1, Amazon.com
27. Quandt family, Germany, 9.9, BMW
28. Mulliez, Gerard, France, 9.8, retail
29. Kwok, Walter, Thomas, Raymond, Hong Kong, 9.7, real estate
30. Ellison, Larry, United States, 9.5, Oracle
31. Rausing, Gad, Sweden, 9.0, packaging
32. Redstone, Sumner, United States, 8.7, Viacom
33. Schwab, Charles, United States, 8.7, discount stock brokerage
34. Berlusconi, Silvio, Italy, 8.0, media
35. Slim Helu, Carlos, Mexico, 8.0, diversified
36. Omidyar, Pierre, United States, 7.8, eBay
37. Takei, Yasuo, Japan, 7.8, consumer finance
38. Turner, Ted, United States, 7.8, Turner Broadcasting
39. Kerkorian, Kirk, United States, 7.3, investments
40. Haefner, Walter, Switzerland, 7.2, computers, cars
41. Murdoch, Rupert, United States, 7.2, publishing
42. Schwarz, Dieter, Germany, 7.1, retailing
43. Boehringer family, Germany, 7.1, pharmaceuticals
44. Olayan, Suliman, Saudi Arabia, 7.1, Olayan Investment Co.
45. Persson, Stefan, Sweden, 7.0, retailing
46. Rausing, Hans, Sweden, 7.0, packaging
47. Beisheim, Otto, Germany, 6.8, retailing
48. Saji, Keizo, Japan, 6.7, Suntory
49. Schmidt-Ruthenbeck family, Germany, 6.6, retailing

Elon Musk is betting Tesla's future isn't about cars at all

Now the company – and its controversial CEO, **Elon Musk** – face stiffer competition and political headwinds. Its **EV sales fell by** a record 9% in 2025, amid increasing rivalry from China and the expiration of the US EV sales tax credit.

But Musk has been steering the company toward an audacious bet. He believes Tesla's future won't ride on cars but on humanoid robots.

On Tesla's earnings call on Wednesday, Musk laid out a literal replacement of Tesla cars by robots – announcing Tesla would discontinue the Model S and Model X in favor of making more of its Optimus robots.

Tesla is not the only company in this space. Hyundai and Google DeepMind are also deploying their Atlas humanoid robot internally in the coming months before rolling it out to customers. Meanwhile the **CES tech show** in January was full of companies — including Nvidia, Qualcomm and Intel — showing off humanoid robots powered by their chips and technologies.

More than 90 companies have a humanoid robot product, according to Ani Kelkar, a partner at McKinsey & Company who leads the firms' advanced automation and autonomy sector. More are coming, especially in the United States and China.

Innovation

The ability to innovate is important for both newly started ventures as well as existing organisations.

Therefore, the concept of innovation has been looked at from a diversity of perspectives and in different disciplines.

This has created a plethora of information about innovation and innovation processes.

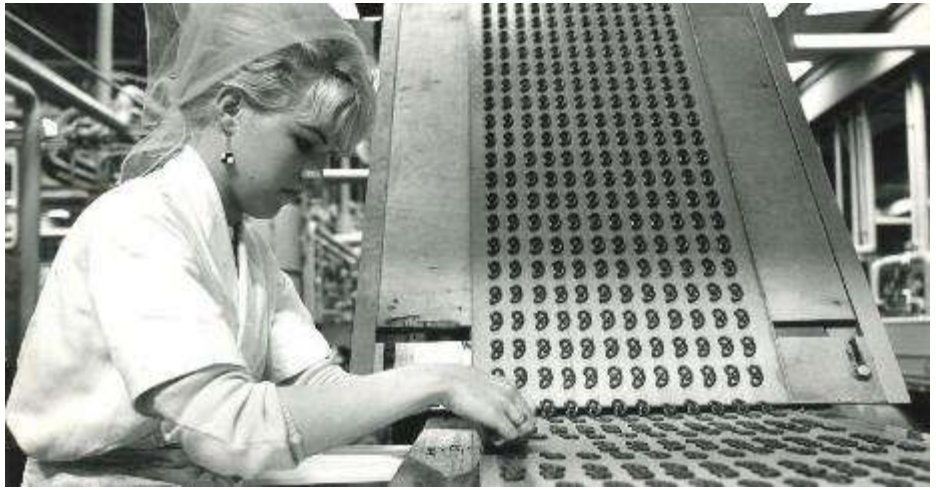
However, it has also resulted in many **myths** about how firms can innovate

Innovation, entreprenuership and business models

- Commonality between the three concepts:
- The Heffalump in Winnie the Pooh
- “The **search** for the **source** of **dynamic entrepreneurial performance** has much in common with hunting the **Heffalump**. The Heffalump is a large and rather important animal (...). He has been hunted by many individuals using various ingenious trapping devices, but no one so far has succeeded in capturing him. All who claim to have caught sight of him report that he is enormous, but they disagree on his particularities.”
  Kilby (1971:275)
- Imagined it looks like this

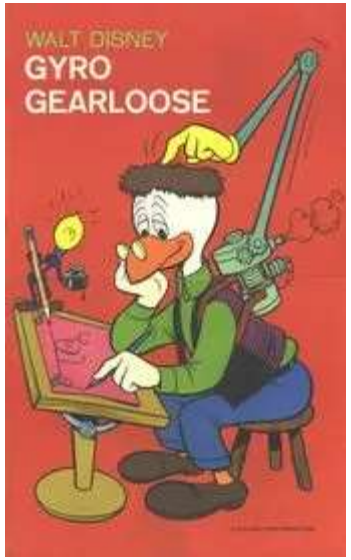
Innovation is not new – case of Toms Group

- 1940s
  - Handmade
  - 15kg person/day
- 1960s
  - 4,000 kg/day
- Today
  - Further automated



Invention versus innovation

- Important to distinguish between invention and innovation
  - “An **invention** is an idea, a sketch or a model for a new or improved device, product, process or system. Such inventions may often (not always) be patented but they do not necessarily lead to technical innovations. In fact the majority do not.
  - **An innovation in the economic sense is accomplished only with the first commercial transaction involving the new product, process system or device**, although the word is used also to describe the whole process.” (Freeman and Soete 1997, p6.)
  - To be able to turn an **invention into an innovation, the innovator normally needs to combine several different types of knowledge, capabilities, skills and resources**. For instance, production knowledge, skills, facilities, market knowledge, a well-functioning distribution system, sufficient financial resources, and so on may be required. Fagerberg, J. (2013). Innovation - a New Guide, TIK WORKING PAPERS on Innovation Studies No. 20131119
  - Innovation sometimes refer to the outcome and/or to the process



New or significantly improved

- **Definition – Oslo manual – Community innovation survey**
  - An innovation is the introduction of a **new or significantly improved** product, process, organisational method, or marketing method by your enterprise.
  - An innovation must have characteristics or intended uses that are new or which provide a significant improvement over what was previously used or sold by **your enterprise**. However, an innovation can fail or take time to prove itself.
  - An innovation need only be new or significantly improved for your enterprise. It could have been **originally developed or used by other enterprises**
- **Innovation** should be seen as relative to something else
- The **economic and social impacts** of inventions and ideas depend on the **diffusion** and uptake of related innovations. (OECD 2018 p44)

- **Novelty – significantly improved**. At a minimum, innovations must contain characteristics that were not previously made available by the relevant organisation to its users.(OECD, 2018)
- These features may or may not be new to the economy, society, or a particular market. An innovation can be based on products and processes that were already in use in other contexts, for instance in other geog ct markets.

Definitions of innovation

Definitions of Innovation

Product innovation. Product innovation is the introduction of a good or service that is new or significantly improved with respect to its characteristics or intended uses. This includes significant improvements in technical specifications, components and materials, incorporated software, user friendliness or other functional characteristics.

Process innovation. Process innovation is the implementation of a new or significantly improved product or delivery method. This includes significant changes in techniques, equipment and/or software.

Marketing innovation. Marketing innovation is the implementation of a new marketing method involving significant changes in product design or packaging, product placement, product promotion or pricing.

Organisational innovation. Organisational innovation is the implementation of a new organisational method in a firm's business practices, workplace organisation or external relations.

Source: OECD (2005b).

Business model innovations

- A business model includes **all core business processes such as the production, logistical, marketing and co-operative arrangements in use as well as the main products that a firm sells, currently or in the future, to achieve its strategic goals and objectives.**
- A firm can use a single business model or several business models at the same time, for instance for different product lines or markets.
- The innovation management literature notes that successful business models combine a method for better meeting the needs of users relative to what competitors can deliver and a profit formula for earning income from delivering utility to customers (Johnson, Christensen and Kagermann, 2008).

Dimensions Novelty

- **Categorisations based on how radical they are/ different from exiting technology**
- **Continuous** improvements are often characterized as **incremental** innovations
- **Radical** innovations (such as the introduction of a totally new type of machinery in a specific industry)
- **Technological revolutions** (consisting of a cluster of innovations that together may have a very far-reaching impact in a whole range of industries or the economy as a whole).
- **Creative destruction**" to characterize the process through which innovation, especially when of the market creating and/or organizational type, "revolutionizes the structure from within, incessantly destroying the old one, incessantly creating the new one." (Schumpeter 1942, p. 83)
- **Disruptive innovation** for innovations that through the exploitation of new markets or market niches gradually undermine the position of existing business models (Christensen1997).
- The **activities of newly created firms** (most of which are service firms) present a potential source of confusion with respect to the basic definition of an innovation because for a period of time a new firm will have no previous products or business processes for comparison. In this case, the comparison group is what is available in the relevant market.(OECD 2018)

Exercise

- Describe this innovation
- Discuss the novelty and potential effect (firm/employees)

Self-service kiosks have expanded at McDonald's and other fast-food chains. Anthony Behar/Sipa/AP

New York (CNN) — Self-service kiosks at McDonald's and other fast-food chains have loomed as job killers since they were first rolled out 25 years ago. But nobody predicted what actually happened.

In one of the earliest mentions of kiosks in fast-food settings in 1999, now-defunct trade industry publication Business Information said that McDonald's was working to "develop an electronic order-taking system that may eventually replace some of its human equivalents."

Definitions of Innovation

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Source: OECD (2005b).

- Three *degrees* of innovation: (i) incremental; (ii) synthetic, and (iii) discontinuous.
- **synthetic** innovation involves the ability to combine existing ideas or technologies in creative ways to produce new products or processes
- **discontinuous** innovation involves radical new ideas that provide breakthrough technologies and advance industries to new levels.

1.12 Types of Innovation

	Product	Process
Incremental	Incremental product change.	Learning by doing.
Synthetic	Dominant designs DC-3, Boeing 707, IBM 360.	Major process improvements. Rotary kiln in cement manufacturing.
Discontinuous	Vacuum tubes to transistors. Piston to jet engines.	Individual wafer to planar process in semiconductor. Float-glass process.

Small

Substantial Learning

Fig. 1.8 Types of innovation. (Source: Tushman and Nadler 1986)

Many definitions – some are less good

Type	Description	Examples
Invention	Totally new product, service or process.	Airplane – Wright bros. Light bulb – Edison. Telephone – A G Bell.
Extension	New use of existing product, service or process.	McDonald's – Kroc. Atari – Bushnell. Holiday Inn – Wilson.
Duplication	Creative replication of existing concept.	Wal-Mart – department store Pizza Hut – pizza parlor.
Synthesis	Combination of existing concepts into new use.	Federal Express – Fred Smith Merrill Lynch – home equity financing.

Fig. 1.9 A typology of innovation. (Source: Kuratko and Hodgetts 2004)

Imitation and diffusion

- **Imitation** is important for innovation management
- Without imitation the social and economic impact of innovation would matter much less (if at all). (Fagerberg p10)
- **introducing** something in a new context often implies **considerable adaptation** (and, hence, marginal innovation) and, as history has shown, organizational changes (or innovations) that may significantly increase productivity and competitiveness.
- Therefore, we need **not only** on how innovations occur, but also on how innovations spread (or diffuse), through imitation or by other means, and the feedback from this process on innovation activity.
- Imitation or improving by learning
  - Learning from competitors
  - Recombination of knowledge
  - Taking existing knowledge from other industries (innovations) and applying it in own firm/industry
  - Innovation is also often a cumulative and continuous process

Product/industry Lifecycles

- Product
  and
- industry lifecycles

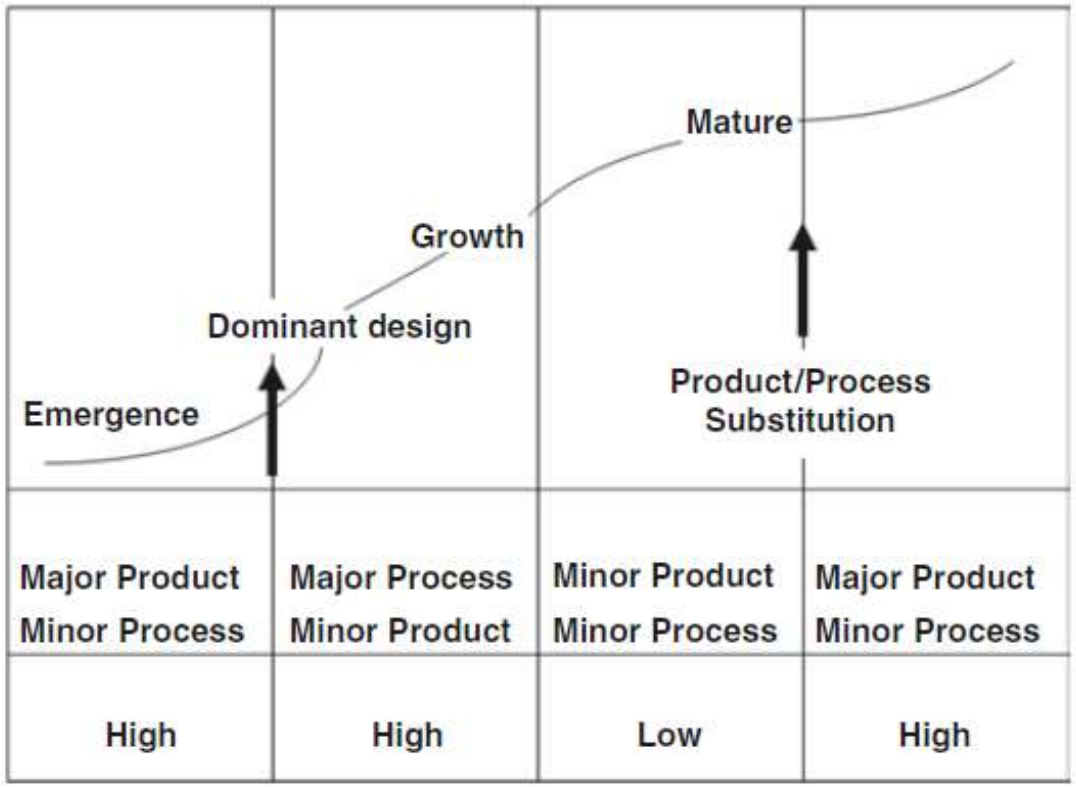


Fig. 1.10 Innovation lifecycles. (Source: Tushman and Nadler 1986)

The role of the national innovation systems

- Firms are also shaped by their location
- And firms shape their location
- **Definition**: “All economic, social, political, organizational, institutional and other factors that influence the development, diffusion and use of innovations” (Edquist, 1997, p.14 or 2003, p.1.)
- While the design of specific NIS varies from country to country, most comprise at least two broad elements.
  - The first is a common innovation infrastructure (e.g. the technological sophistication of the country, investment in R&D, government support for innovation).
  - The second is the cluster specific environment for innovation (e.g. competitive rivalry, ease of exit and entry, supplier/buyer power) found within the industries in which the firms are operating (Porter and Stern 2001).

A brief intro to innovation systems

- Systems of innovation can be defined as a narrow concept that focuses on science-based learning and firms' science-–technology-–innovation (STI) activities, or as a broader concept that also includes doing-–using-–interacting (DUI) learning in the economy.
- The broader definition involves an understanding that **knowledge** is the most important resource in an economy and that **learning** is a critical **process** in creating, diffusing, and utilizing the knowledge .
- Innovation ecosystems
  - Firms increasingly form 'innovation ecosystems' to implement **complex value propositions** Defined as "**the collaborative arrangements through which firms combine their individual offerings into a coherent, customer-facing solution** (Adner 2006)", at the core of an innovation ecosystem, one often finds a technology platform: a set of shared assets, standards, and interfaces that underpins an activity system surrounding it (Dattée, Alexy and Autio (2018)

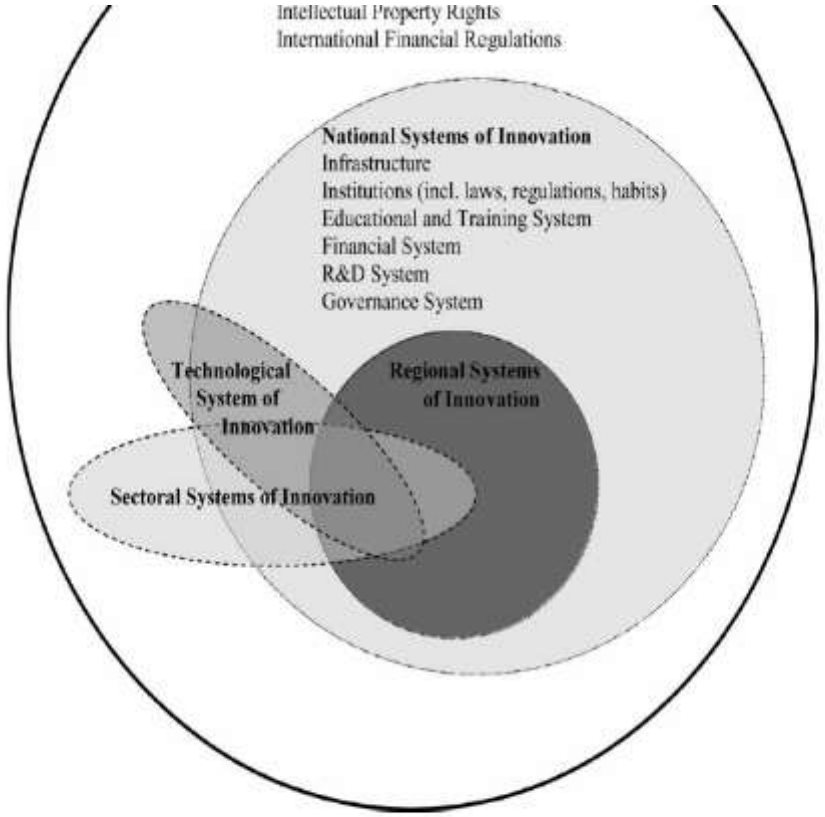
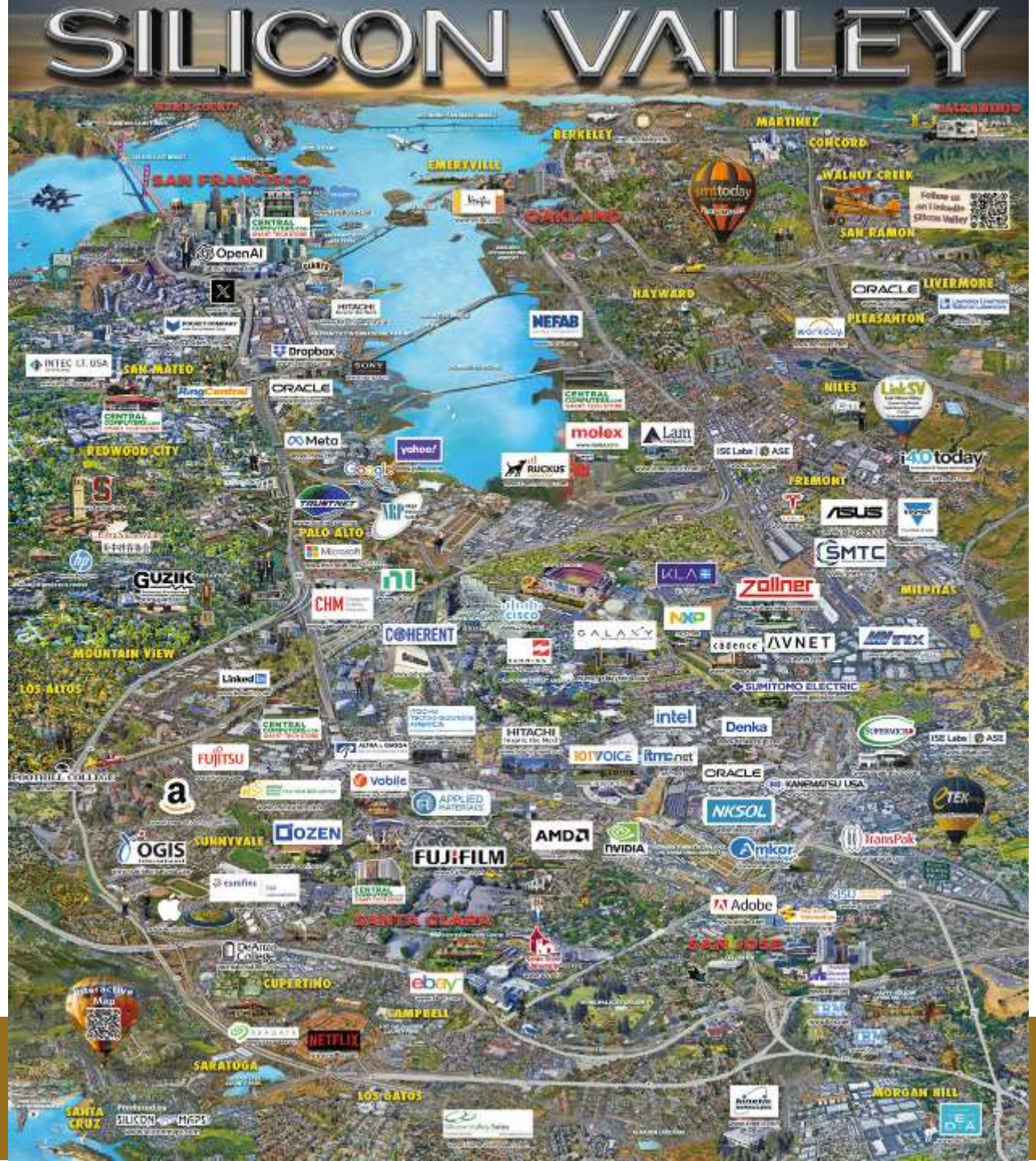


Fig. 3. Relationship between global, national, regional, sectoral and technological systems of innovation
Source: FRENZ and OUGHTON (2005)


SILICON VALLEY
SAN FRANCISCO
EMERYVILLE
BERKELEY
OAKLAND
MARTINEZ
CONCORD
WALNUT CREEK
SAN RAMON
HAYWARD
LIVERMORE
PLEASANTON
NILES
FREMONT
MILPITAS
SAN MATEO
REDWOOD CITY
PALO ALTO
MOUNTAIN VIEW
LOS ALTOS
SUNNYVALE
SANTA CLARA
SAN JOSE
CUPERTINO
CAMPBELL
SARATOGA
LOS GATOS
MORGAN HILL
SANTA CRUZ
OpenAI
Dropbox
ORACLE
Meta
yahoo!
Google
NEFAB
molex
Lam
RUCKUS
ASUS
SMTC
Zollner
KLA
NXP
cadence
AVNET
SUMITOMO ELECTRIC
intel
Denka
HITACHI
IOTVOICE
itmc.net
KANEMATSU USA
NKSOL
AMD
NVIDIA
Amkor
TransPak
FUJIFILM
APPLIED MATERIALS
Vobile
OZEN
OGIS
FUJITSU
Linked
COHERENT
CHM
GUZIK
Adobe
ebay
NETFLIX
Microsoft
Apple
Amazon
IBM
Rinetu
EDA
i40today
smttoday
Follow us on LinkedIn Silicon Valley
Interactive Map
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Diversity in Innovation

- Types of Innovation:
  - Diverse forms of innovation exist, ranging from product and process innovations to business model and organizational innovations.
- Degrees of Novelty:
  - Innovations can vary in their degree of novelty, from incremental improvements to radical, game-changing breakthroughs.
- Impact Variability:
  - The impact of innovations can differ significantly, influencing market dynamics, customer experiences, and industry landscapes.
- Sources of Innovation:
  - Innovations can stem from various sources, including internal research and development, collaborations, and external partnerships.
- Industry and Firm Specifics:
  - Different industries and firms may employ distinct innovation strategies tailored to their unique challenges and opportunities.
- Stages in the Industry Life Cycle:
  - The industry life cycle encompasses conception, development, implementation, and diffusion stages, each presenting distinct challenges and considerations.
- Contextual Variances:
  - Innovation is shaped by contextual factors such as technologies, industries, regions, and national policies, leading to a rich tapestry of approaches.
- **Despite this diversity, certain stable patterns and approaches in innovation persist, providing a foundation for understanding and navigating the innovation landscape.**

Innovation is a process

Idea → Innovation

time

- Innovation is not a linear, sequential process, but involves many **interactions** and **feedbacks** in knowledge creation and use.
- In addition, **innovation is based on a learning process that draws on multiple inputs and requires ongoing problem-solving**.
- The systems perspective of innovation calls for multidisciplinary and interdisciplinary approaches to examine the interdependencies among actors, the **uncertainty** of outcomes, as well as the **path-dependent** and **evolutionary** features of systems that are **complex** and **non-linear** in their responses to policy intervention.

Innovation and entrepreneurship

- Entrepreneurship tend to focus on the entrepreneur
- Innovation does not focus on the inventor, but on the innovator

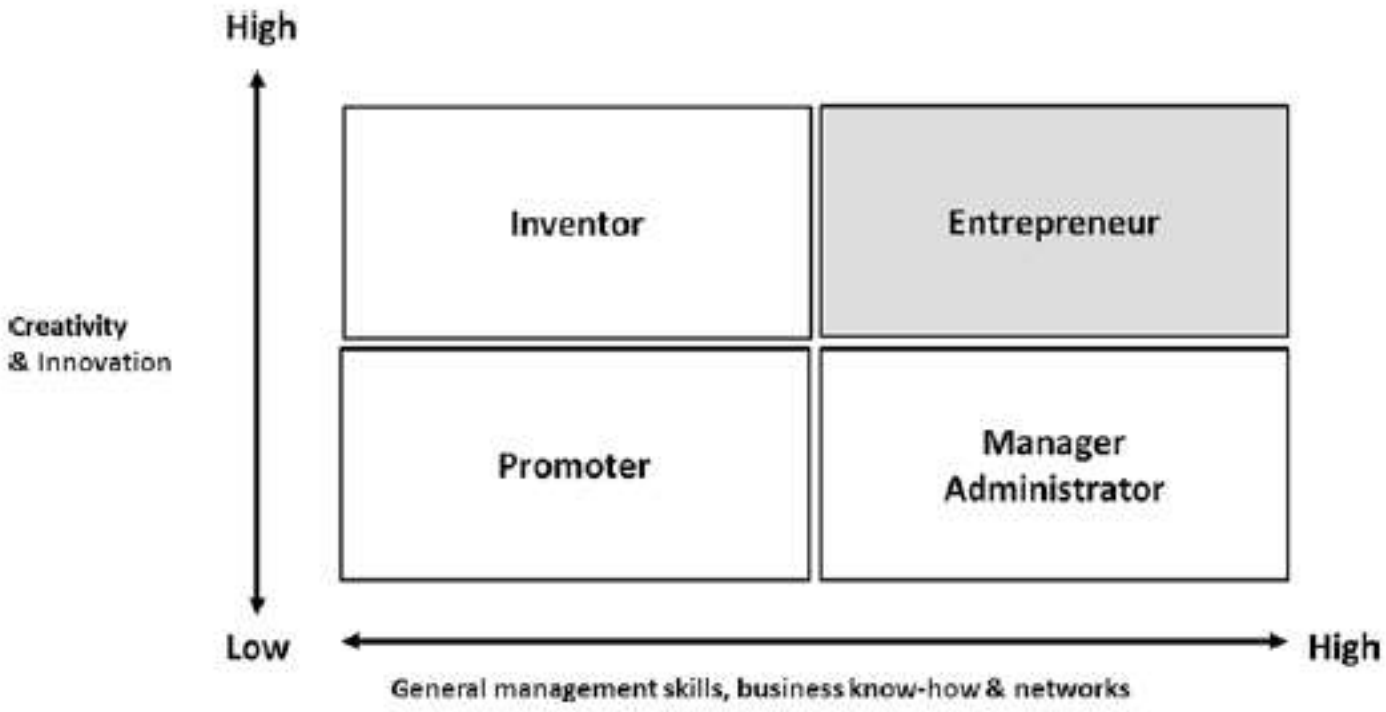

Fig. 1.5 Who is the real entrepreneur? (Source: Timmons 1999)

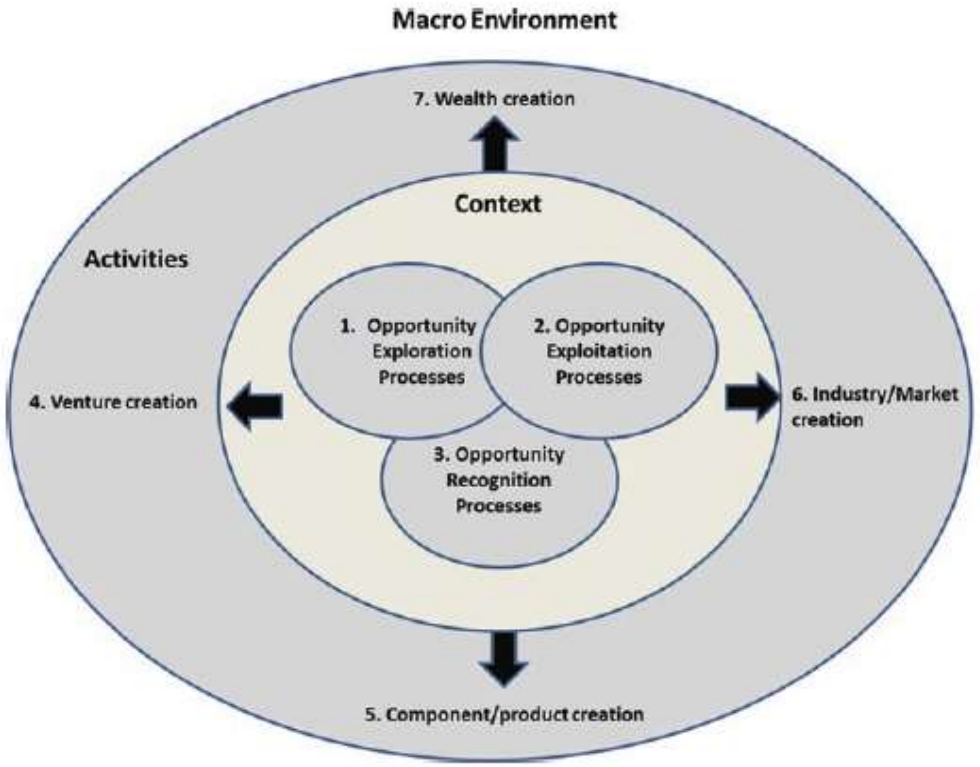

Fig. 1.4 The entrepreneurship domain. (Source: Brush et al. 2003)

So… what do we know – and what do we need to know

- Innovation is important
  - Competitive sword
  - Innovation is necessary for firms' survival
- But it is a difficult process
- Crisis response vs. planned process vs. innovative or entrepreneurial organization
- Incremental innovations → radical innovations
- However, we still need to know
  - what different firm do to innovate
  - The context for innovation, diffusion and adoption
  - How to deal with structure and behavior in innovation processes

What have you learned today

- What is an innovation
  - Entrepreneurship and innovation
  - Innovation relative to something else
  - Innovation can be characterized along different dimensions
  - Innovation patterns changes over time in industries
  - Innovation process is not linear
  - Innovation involves several distinct phases
  - Innovation as an interactive process
  - The role of the national system of innovation